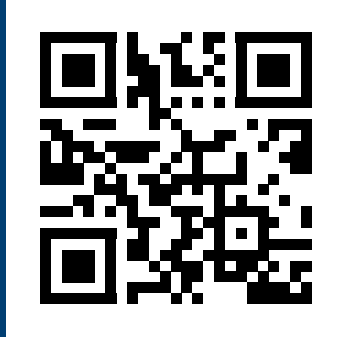




Digitization of Historical Meteorology Graphs from the Kandilli Observatory Archive

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GitHub Repository

Motivation

Between **1940 and 1990**, the **Kandilli Observatory**'s temperature record survives almost only as *thermograms* — continuous ink traces drawn by a mechanical pen on a rotating drum chart. The archive holds roughly **1,682 TIFF scans**, but the time series they represent has never been captured in machine-readable form — so it cannot be used in climate reanalysis, trend studies, or modern visualizations. Manual transcription would be a multi-year effort.

Why It Is Hard

Three properties of the data defeat a naive automated approach:

- **The curve is tiny.** The ink trace is only 0.1–0.5 % of the pixels, drawn over a dominant printed grid it crosses hundreds of times.
- **The grid is curved.** Printing on a cylindrical drum bends the vertical grid lines into a barrel-distorted parabola, so the time axis is not straight.
- **The archive is heterogeneous.** Five decades and three chart formats (Daily, 4-day, Weekly) yield **nine visually distinct templates** differing in ink and grid color (orange, green, yellow, pink).

A single fixed pipeline cannot handle all of them.

Goal & Outcome

A **cross-platform desktop application** (macOS + Windows) that turns each scanned thermogram into a clean (**datetime**, **temperature °C**) CSV, ready for analysis.

- **Human-in-the-loop:** an image-processing pipeline produces an automatic extraction; a focused editing UI lets a researcher correct it.
- **No programming needed** — designed for a domain expert.
- **Runs fully offline** as a single bundled binary.
- **Provenance preserved:** every exported point keeps its source pixel coordinates and a flag marking manual edits.

Why Classical Computer Vision?

We deliberately avoided a learned segmentation model [5]: no labeled training data exists, the archive is too heterogeneous to annotate exhaustively, and a **transparent** pipeline can be inspected, calibrated, and corrected by a domain expert — which a black-box network cannot. The system instead relies on **per-template color and geometry profiles**, selected automatically when a scan is uploaded.

Broad Impact

A continuous, structured daily temperature record for the second half of the twentieth century is a building block for **climate trend analysis**, validation of reanalysis products, and downscaling of regional climate models over the Eastern Mediterranean.

The workflow also generalizes: the same per-template, human-in-the-loop approach transfers directly to the paper archives held by other **national meteorological services and data-rescue projects**. At the scale of 1,682 charts, reducing transcription from minutes per chart to **well under a minute** is the difference between a feasible project and a multi-year manual effort.

Ethical Considerations

Raw scans remain the property of the Kandilli Observatory and are never redistributed; the app only writes derivative CSVs next to the user's own images. The data contains no personal information — only air temperature. To prevent over-trusting an automated series, every point retains its source pixel coordinates, edited points are flagged in the export, and extraction requires explicit human calibration first.

System Architecture

Three cooperating processes: the **React** frontend renders the workspace and overlays; the **Rust** / **Tauri** [1] host exposes the IPC commands and owns file I/O; the **Python** sidecar runs the pipeline on OpenCV [2] and NumPy [3]. Every trip goes through the Rust host.

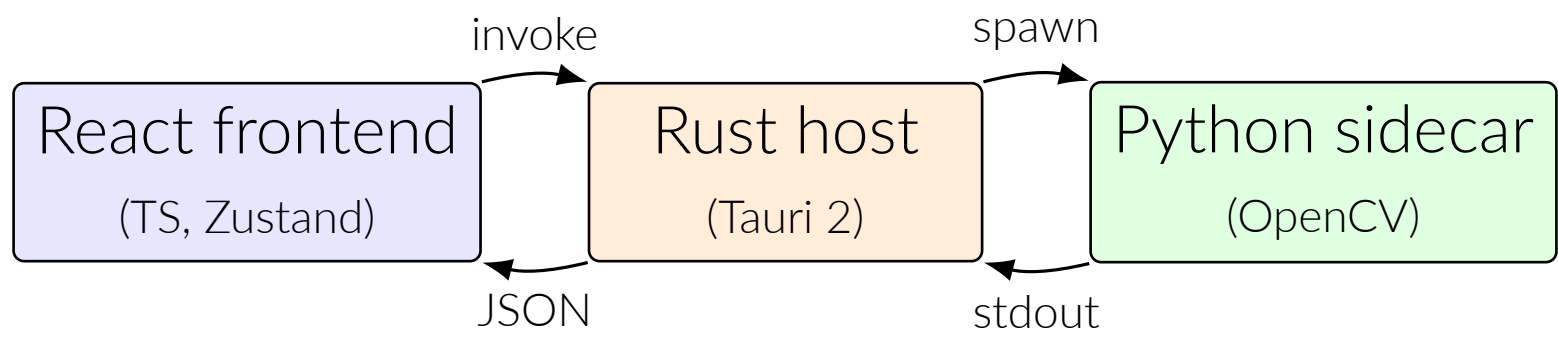


Figure 1. Three-process architecture. Long-lived state is written to disk as per-template calibration JSON and CSV files.

Processing Pipeline

Each scan flows through six stages with one classifier at the front; each stage hands a typed result to the next.

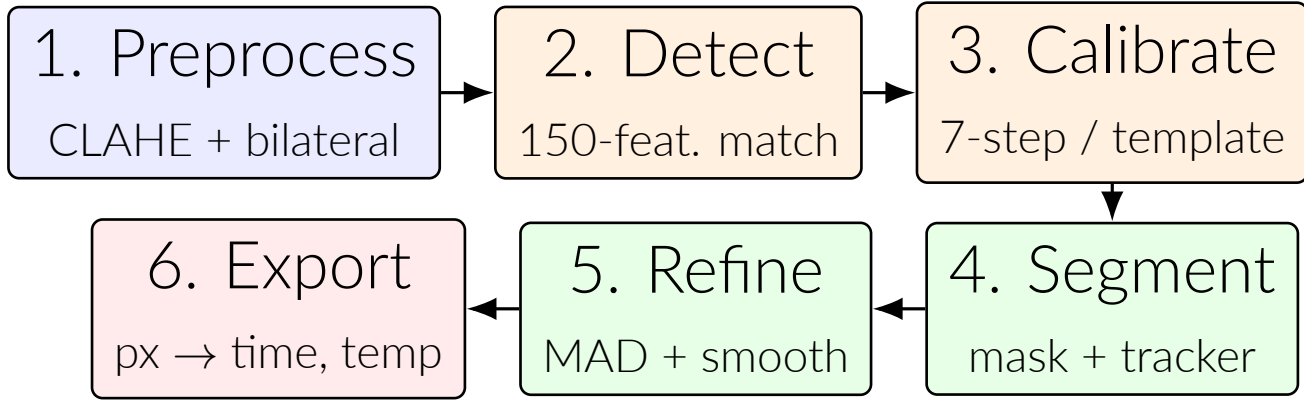


Figure 2. Pipeline. Blue runs on upload; orange once per template; green on every curve request; red is user-driven.

Sliding-window tracker (core idea)

A binary curve mask from per-template RGB rules is walked column-by-column, taking the *median* of mask pixels in a tight band around an expected *y* and capping the per-column jump — so one algorithm follows a curve that crosses the grid hundreds of times without being pulled off by grid lines, stamps, or handwriting. Outliers are cut with MAD [4]; drum distortion is modeled as $x = x_{\text{ref}} + a(y - c_y)^2$ and inverted on export to recover the true time.

Guided Calibration

A **seven-step** modal fixes rotation, temperature, and grid spacing once per template, saved as JSON and reused on every later scan.

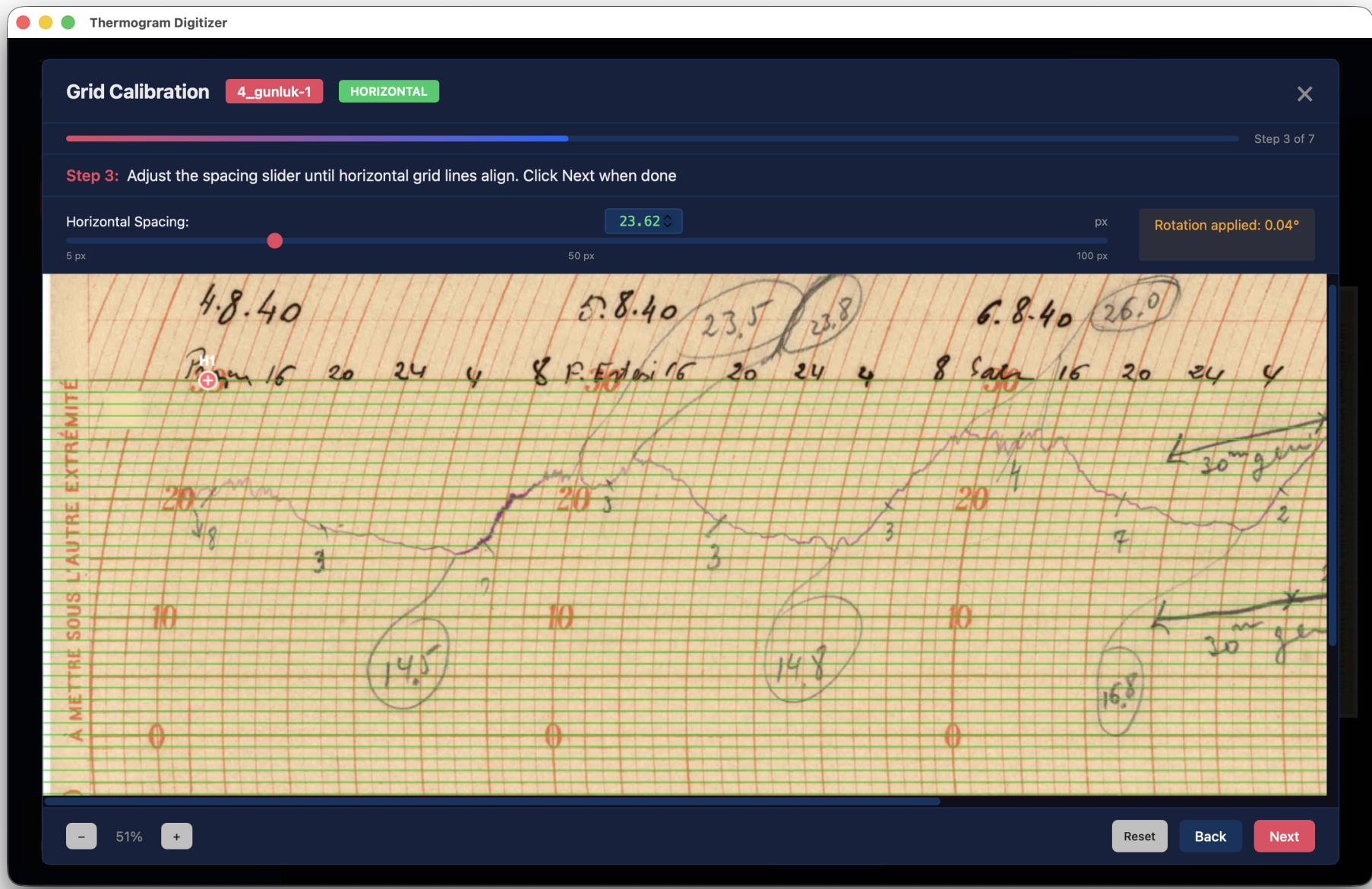


Figure 3. Calibration step 3 of 7: aligning grid spacing on the live overlay.

The Application

Switching to *Curve* view extracts and overlays the trace automatically; damaged regions are fixed in seconds by setting start/end points, dragging or deleting points, freehand-drawing a correction that snaps to the ink, or re-extracting a region — all with full undo / redo.

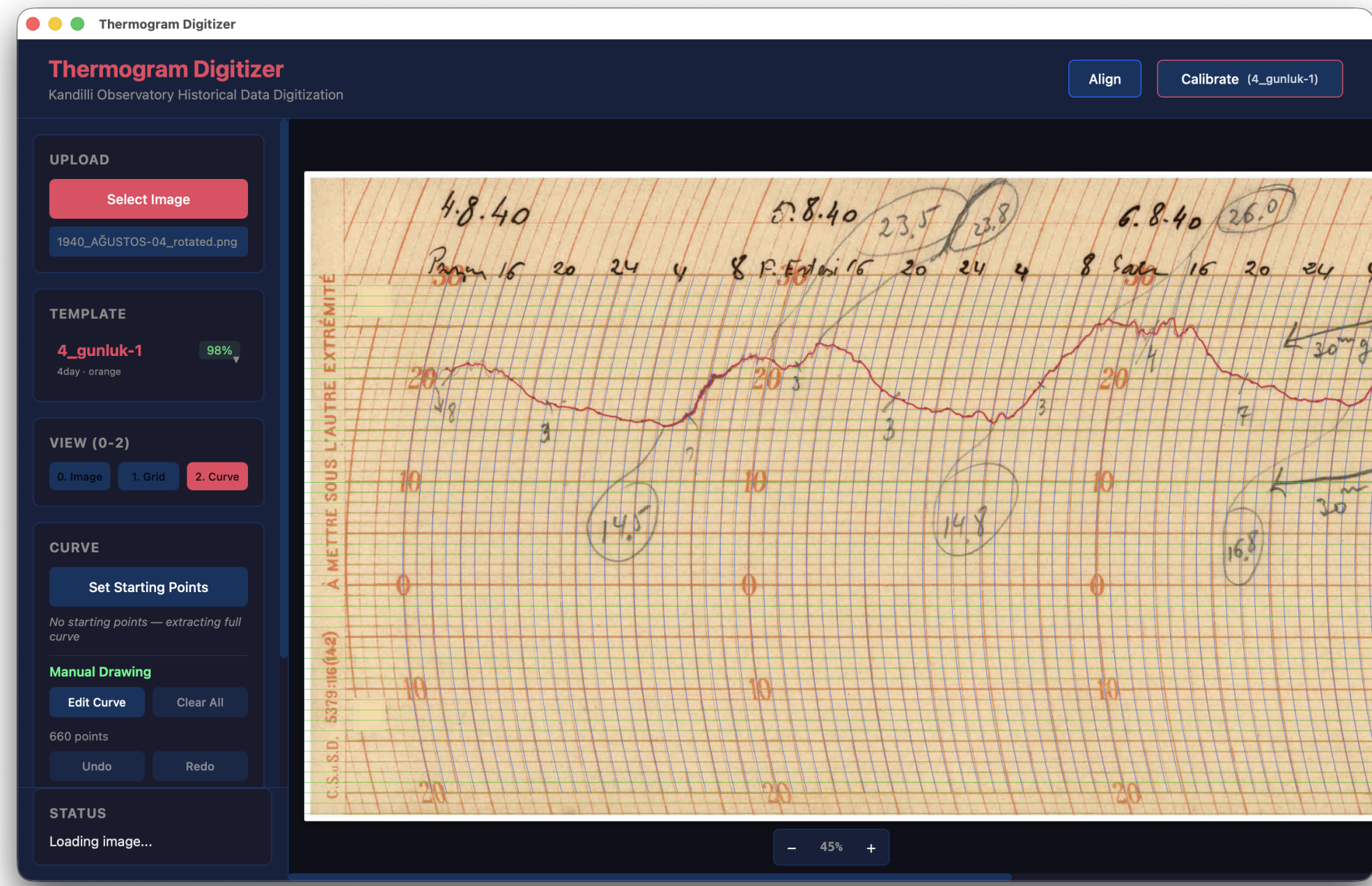


Figure 4. Automatic extraction (red) on a clean 1940 four-day scan. Template **4_gunluk-1** detected at 98 %.

Deployment

The application ships as a single platform-specific binary built with **cargo tauri build** — a .app bundle on macOS and an MSI installer on Windows. The Python interpreter and backend source are bundled inside, so the researcher installs once and launches the sidecar on demand with **no virtualenv, no PATH setup, and no network access** required. All grid geometry lives in flat per-template JSON files, keeping the system inspectable and trivially portable between machines.

Results

Across the project dataset, the desired output is reached with little or no manual work:

- 87 % — only start/end point marking, **zero** editing operations.
- 9 % — a single refinement / fixation.
- 4 % — two or more refinements.

So roughly **nine out of ten charts** are digitized without any curve editing at all.

- < 3 s per extraction on a laptop with no GPU.

Conclusion

Classical computer vision — per-template color profiles plus a cylindrical-grid calibration — reaches a quality where most charts need no editing and the rest need only a few corrections. A transparent, auditable pipeline proved the right trade-off and transfers directly to other paper climate archives [6].

References

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[2] G. Bradski, *The OpenCV Library*. Dr. Dobb's Journal, 2000.
[3] C. R. Harris et al. *Array programming with NumPy*. Nature 585, 2020.
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[6] S. Brönnimann et al. *Unlocking pre-1850 instrumental meteorological records*. BAMS 100, 2019.